



TF

HORMONES

Natural Testosterone Optimization

Lifestyle + Nutrition Science

48 Pages

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Chapter 1: How Testosterone is Produced — The HPG Axis

Hypothalamus releases GnRH (Gonadotropin-Releasing Hormone) [Triggered by: sleep, exercise, nutrition, stress levels]



GnRH signals the Pituitary Gland to release LH (Luteinizing Hormone) + FSH (Follicle-Stimulating Hormone)



LH travels through bloodstream to the Testes and stimulates Leydig cells to produce Testosterone



Testosterone enters bloodstream Free T (~2%) + SHBG-bound T (~44%) + Albumin-bound T (~54%)



Negative Feedback Loop: High T suppresses GnRH + LH Estrogen (from aromatisation) also inhibits the axis



Target: Total T 600–900 ng/dL | Free T 15–25 pg/mL Maintain with lifestyle, nutrition, sleep

Blood Marker	Optimal Range	Low Indicates	High Indicates
Total Testosterone	600–900 ng/dL	Hypogonadism, lifestyle factors	Exogenous T use (if very high)
Free Testosterone	15–25 pg/mL	High SHBG, low total T	Low SHBG
SHBG	20–50 nmol/L	Obesity, insulin resistance	Ageing, hyperthyroidism
LH (Luteinizing H.)	2–8 IU/L	Pituitary/hypothalamic issue	Primary hypogonadism
FSH	1.5–12 IU/L	Pituitary suppression	Testicular failure
Estradiol (E2)	20–30 pg/mL	Bone loss risk	Gynecomastia, libido issues
DHT	30–85 ng/dL	Low T or 5-AR inhibition	Hair loss, prostate risk
Prolactin	<15 ng/mL	Normal	Pituitary adenoma, T suppression

Chapter 2: Top T-Boosting Foods — Nutrient Analysis

Food	Key Nutrient	Amount/100g	T-Support Mechanism	Recommended Intake
Oysters	Zinc	78mg (711% DV)	Zinc essential for LH synthesis + T production	2–3 pieces/week
Pumpkin seeds	Zinc + Mg	7.7mg Zn, 592mg Mg	Both nutrients directly support T synthesis	30g/day
Eggs (whole)	Cholesterol + D	185mg chol, 1.1mcg D3	Cholesterol = T precursor; D3 boosts T	3–4 whole eggs/day
Beef (lean)	Zinc + Saturated fat	4.5mg Zn/100g	Dietary fat needed for steroidogenesis	150g 3x/week
Fatty fish (Salmon)	Vitamin D + Omega-3	526 IU D3/100g	Vitamin D receptor found on Leydig cells	150g 2–3x/week
Ashwagandha	Withanolides	Standardised 5%	↑ LH, ↑ T by 15–40% in studies	600mg extract/day
Garlic (raw)	Allicin + DATS	High allicin	↓ Cortisol (which suppresses T)	3 cloves/day raw
Pomegranate	Punicalagins	High polyphenols	↑ salivary T 24% in one study	1 glass juice/day
Avocado	Boron + MUFA	1mg boron/100g	Boron reduces SHBG, increases free T	½ avocado/day
Brazil nuts	Selenium	544mcg/100g	Selenium needed for testicular function	2–3 nuts/day (max)
Dark chocolate (85%+)	Magnesium + Zinc	176mg Mg/100g	Mg reduces SHBG, frees up testosterone	20–30g/day
Fenugreek (Methi)	4-HO-Isoleucine	Seed extract	Inhibits 5-AR and aromatase (natural)	500–600mg/day

Chapter 3: Sleep Quality vs Testosterone — The Critical Link

70% of daily testosterone release occurs during sleep, specifically during REM sleep phases. Even one week of sleeping 5 hours per night reduces testosterone levels by 10–15% in young healthy men (University of Chicago,

2011).

Sleep Duration	T Level Impact	GH Release	Cortisol Next Day	Action
<5 hours	↓ 10–15%/week	↓ 50%	↑ 37%	Critical — fix immediately
5–6 hours	↓ 5–8%	↓ 30%	↑ 18%	Below optimal
6–7 hours	↓ 2–3%	↓ 10%	↑ 8%	Acceptable, not ideal
7–8 hours	Baseline/Optimal	Peak release 11pm–3am	Normal	■ Target range
8–9 hours	+3–5% vs 7h	Full cycle achieved	↓ 5%	■ Ideal for athletes
9+ hours	Minimal extra benefit	Similar to 8–9h	Normal	OK if needed, diminishing returns

Chapter 4: Training Frequency vs Testosterone Response

Resistance training causes acute spikes in testosterone (30–45 min post-session). However, excessive training volume without adequate recovery chronically suppresses T through cortisol elevation and HPA axis overactivation.

Training Protocol	Acute T Spike	Chronic T Effect	Optimal For
Heavy compound lifts (≥80% 1RM)	↑ 15–25%	↑ 5–10% baseline	T optimisation
High volume hypertrophy (moderate %)	↑ 10–20%	↑ 3–8% baseline	Muscle + T
Short rest periods (<60s)	↑ 20–30%	Neutral to slight ↑	Metabolic training
Long rest (2–3 min)	↑ 10–15%	↑ strength, neutral T	Strength focus
Overtraining (>6 days/week high V)	↑ initially then ↓	↓ 5–25% baseline	Avoid

Training Protocol	Acute T Spike	Chronic T Effect	Optimal For
Cardio only (no resistance)	Minimal ↑	↓ or neutral long term	Not ideal for T
No exercise (sedentary)	None	↓ over months	Avoid — use it or lose it

Chapter 5: Supplement Evidence Table

Supplement	Dosage	Evidence Level	T Effect	Notes
Ashwagandha (KSM-66)	600mg/day	★★★★★ (A)	↑ 14–40%	Best studied, safe, reduces cortisol too
Vitamin D3	2,000–5,000 IU/day	★★★★ (A)	↑ 25% in deficient men	Check baseline serum D3 first
Zinc	25–45mg/day	★★★★ (A)	↑ 8–24% if deficient	Zinc Picolinate best absorbed form
Magnesium Glycinate	400–600mg/day	★★★★ (A)	↑ free T via ↓SHBG	Take at night — improves sleep too
Fenugreek extract	500–600mg/day	★★★★ (B+)	↑ 6–12%	Inhibits 5-AR, may reduce DHT slightly
Boron	10mg/day	★★★ (B)	↑ free T via ↓SHBG	Reduces SHBG by 9% at 10mg
Tongkat Ali (100:1 extract)	200–400mg/day	★★★ (B)	↑ 8–15%	Reduces SHBG, cortisol, improves LH
DHEA	25–50mg/day	★★★ (B)	Converts to T in body	Only if DHEA-S is low on blood test
L-Carnitine L-Tartrate	2g/day	★★★ (B)	↑ androgen receptor density	Supports T action in muscle
D-Aspartic Acid	3g/day	★★ (C)	↑ short-term in untrained	Loses effect after 4–6 weeks
Tribulus Terrestris	750mg/day	★ (D)	No significant human data	Overhyped — skip

Supplement	Dosage	Evidence Level	T Effect	Notes
Maca Root	1.5–3g/day	★★ (C)	↑ libido, not T directly	Use for libido, mood, energy

Chapter 6: 90-Day Testosterone Optimization Protocol

Phase	Weeks	Key Actions	Expected Improvement
Foundation	1–4	Fix sleep (7–8h), start D3+Mg+Zinc, 3x strength training	+5–10% T baseline
Nutrition Optimisation	5–8	Add ashwagandha + fenugreek, increase dietary fats, reduce alcohol completely	+10–20% additional
Training Intensification	9–12	Heavy compound focus (squat, DL, bench, OHP), add HIIT 2x/week	+8–15% from training
Lifestyle Refinement	13–16	Stress protocols, sun exposure 20 min/day, cold showers AM	+5–10% from lifestyle
Measurement & Adjust	17–20	Retest blood panel, adjust supplements, maintain all habits	Sustained elevation
Maintenance	21–90	All habits maintained, cycling ashwagandha (8 weeks on, 4 off)	25–40% above baseline